

Extending Quadratic Scaling Laws to Deep NTKs

Via Finite-Width Correction Tensors

Research Strategy Note

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Abstract

This document outlines a rigorous strategy to extend the spectral analysis of the Neural Tangent Kernel (NTK) in the quadratic scaling regime ($n \asymp d^2$) from two-layer networks (Benigni-Paquette, 2025) to multi-layer networks. The core technical mechanism relies on the **Gaussian Equivalence Principle (GEP)**. We verify GEP by mapping the *Tensor* T in the Benigni-Paquette framework to the $O(1/p)$ connected four-point functions (cumulants) established in finite-width network literature.

1 The Setting and The Challenge

1.1 Scaling Regime

We consider a fully-connected deep network with layers $\ell = 0, 1, \dots, L$ and widths d_ℓ . The data matrix is $X^{(0)} \in \mathbb{R}^{n \times d_0}$ with i.i.d. entries (centered, unit variance, finite fourth moment). For $\ell \geq 1$ we define the pre-activations and activations

$$Z^{(\ell)} := \frac{1}{\sqrt{d_{\ell-1}}} X^{(\ell-1)} W^{(\ell)} \in \mathbb{R}^{n \times d_\ell}, \quad X^{(\ell)} := \sigma(Z^{(\ell)}) \in \mathbb{R}^{n \times d_\ell}, \quad (1)$$

where $W^{(\ell)} \in \mathbb{R}^{d_{\ell-1} \times d_\ell}$ has i.i.d. $\mathcal{N}(0, 1)$ entries, independent across layers.

The quadratic scaling regime is:

$$\begin{aligned} n &\rightarrow \infty, \quad d_l \rightarrow \infty \\ \frac{n}{d_0 d_1} &\rightarrow \gamma_1 \in (0, \infty) \quad (\text{Quadratic Samples}) \\ \frac{d_l}{d_{l-1}} &\rightarrow \gamma_{2,l} \in (0, \infty) \quad (\text{Linear Width Ratios}) \end{aligned}$$

In particular $n \asymp d_0^2$ and every width is only *linear* in d_0 . Every single Gram matrix of the form $\frac{1}{d_\ell} X^{(\ell)} X^{(\ell)\top}$ is low-rank (rank at most $d_\ell \asymp d_0$), but Hadamard products of such low-rank factors can have rank up to the product of ranks, which is of order n in this regime.

1.2 The Recursive Challenge

The NTK at layer L , denoted $\Theta^{(L)}$, follows the recursion:

$$\Theta^{(L)} = \Theta^{(L-1)} \odot \dot{\Sigma}^{(L)} + \Sigma^{(L)} \quad (2)$$

The Benigni-Paquette (BP) result characterizes the spectrum of matrices of the form $K = (XX^\top) \odot M$. To apply this recursively, we must treat the hidden layer activations $X^{(l)}$ as the “input data” for layer $l + 1$. **Problem:** $X^{(l)}$ is not i.i.d. Gaussian. Its columns are correlated.

1.3 What we keep from BP (the proof spine)

The BP proof has three reusable steps.

1. A replacement argument that turns a nonlinear feature matrix into “linear + independent noise” without changing the ESD.
2. A tensorization step that rewrites a Hadamard product of Gram matrices as an ordinary Gram matrix of tensor-product features.
3. A Bai–Zhou type theorem: if the row features are independent and their covariance tensor has a deterministic limit, then the ESD is given by a Marchenko–Pastur map (free multiplicative convolution with MP).

To extend BP to deep networks, we need a version of Step 1 that works for the deep feature maps $X^{(\ell)}$, and a way to control the covariance tensor in Step 3 through known finite-width cumulant formulas.

2 Deep scalar-output NTK at initialization

2.1 Model

Fix a depth $L \geq 1$. Let $X^{(0)} \in \mathbb{R}^{n \times d_0}$ be the data matrix with i.i.d. entries. For each layer $\ell = 1, \dots, L$, let $W^{(\ell)} \in \mathbb{R}^{d_{\ell-1} \times d_\ell}$ have i.i.d. $\mathcal{N}(0, 1)$ entries, independent across layers and independent of $X^{(0)}$. Let $a \in \mathbb{R}^{d_L}$ have i.i.d. entries, centered, bounded, and independent of all $W^{(\ell)}$. Define

$$Z^{(\ell)} := \frac{1}{\sqrt{d_{\ell-1}}} X^{(\ell-1)} W^{(\ell)}, \quad X^{(\ell)} := \sigma(Z^{(\ell)}),$$

where σ is applied entrywise. We consider the scalar-output network

$$f(X^{(0)}) := \frac{1}{\sqrt{d_L}} X^{(L)} a \in \mathbb{R}^n. \quad (3)$$

2.2 Backpropagation features

Let $\phi := \sigma'$ (defined Lebesgue-a.e. and applied entrywise). Define the backpropagated signals $\Delta^{(\ell)} \in \mathbb{R}^{n \times d_\ell}$ by the recursion

$$\Delta^{(L)} := \phi(Z^{(L)}) \odot (\mathbf{1} a^\top), \quad (4)$$

$$\Delta^{(\ell)} := \phi(Z^{(\ell)}) \odot \frac{1}{\sqrt{d_\ell}} \Delta^{(\ell+1)} W^{(\ell+1)\top}, \quad \ell = L - 1, \dots, 1, \quad (5)$$

where $\mathbf{1} \in \mathbb{R}^n$ is the all-ones vector and $\odot$ is the Hadamard product.

2.3 Decomposition of the empirical NTK

Let $\Theta^{(L)} \in \mathbb{R}^{n \times n}$ be the empirical NTK at initialization,

$$\Theta^{(L)} := \nabla_{\theta} f(X^{(0)}) \nabla_{\theta} f(X^{(0)})^{\top},$$

where θ collects all weights $\{W^{(\ell)}\}_{\ell=1}^L$ and the output weights a . Then $\Theta^{(L)}$ splits as a sum of a conjugate-kernel term and L Hadamard terms:

$$\Theta^{(L)} = K^{\text{CK}} + \sum_{\ell=1}^L K^{(\ell)}, \quad (6)$$

where

$$K^{\text{CK}} := \frac{1}{d_L} X^{(L)} X^{(L)\top}, \quad (7)$$

and for $\ell = 1, \dots, L$,

$$K^{(\ell)} := \frac{1}{d_{\ell-1}} X^{(\ell-1)} X^{(\ell-1)\top} \odot \frac{1}{d_{\ell}} \Delta^{(\ell)} \Delta^{(\ell)\top}. \quad (8)$$

Remark 1 (Low rank of the conjugate kernel). *We have $\text{rank}(K^{\text{CK}}) \leq d_L$. Under the quadratic scaling, $d_L/n \rightarrow 0$, hence K^{CK} does not affect the bulk ESD.*

3 Deep NTK as a sample covariance in a tensor-product feature space

3.1 Layerwise Jacobian factorization

For a standard scalar-output MLP in NTK parametrization, the Jacobian with respect to the weights in layer ℓ has the outer-product form

$$\nabla_{W^{(\ell)}} f(x_i) \propto \delta^{(\ell)}(x_i) \otimes x_i^{(\ell-1)},$$

where $x_i^{(\ell-1)} \in \mathbb{R}^{d_{\ell-1}}$ is the $(\ell-1)$ -st layer activation vector for sample i , and $\delta^{(\ell)}(x_i) \in \mathbb{R}^{d_{\ell}}$ is the backpropagated “error” at layer ℓ (a product of upper-layer weights and derivatives). Consequently, the layer- ℓ contribution to the NTK has the Hadamard form

$$K^{(\ell)} = \frac{1}{d_{\ell-1}} X^{(\ell-1)} X^{(\ell-1)\top} \odot \frac{1}{d_{\ell}} \Delta^{(\ell)} \Delta^{(\ell)\top}, \quad (9)$$

where $\Delta^{(\ell)} \in \mathbb{R}^{n \times d_{\ell}}$ is the matrix whose i -th row is $\delta^{(\ell)}(x_i)^{\top}$.

Remark 2 (What matches BP exactly, and what is new in depth). *For $\ell = L$ (the last hidden layer in the scalar-output model), the backprop signal has the BP form. Indeed, from (4) we have*

$$\Delta^{(L)} = \phi(Z^{(L)}) \text{Diag}(a),$$

so

$$\frac{1}{d_L} \Delta^{(L)} \Delta^{(L)\top} = \frac{1}{d_L} \phi(Z^{(L)}) \text{Diag}(a)^2 \phi(Z^{(L)})^{\top},$$

which is exactly the “derivative Gram with a fixed diagonal weight” appearing in the two-layer BP model.

For intermediate layers $\ell < L$, the factor multiplying $\phi(Z^{(\ell)})$ is sample-dependent:

$$\Delta^{(\ell)} = \phi(Z^{(\ell)}) \odot U^{(\ell)}, \quad U^{(\ell)} := \frac{1}{\sqrt{d_\ell}} \Delta^{(\ell+1)} W^{(\ell+1)\top}.$$

Hence $\Delta^{(\ell)} \Delta^{(\ell)\top}$ is not of the simple form $Y D^2 Y^\top$ with a fixed diagonal D . This sample-dependence is the main new obstacle in extending BP to depth: one must control the joint fluctuations of the pair $(X^{(\ell-1)}, \Delta^{(\ell)})$ beyond the two-layer setting.

Remark 3 (Low rank versus full rank). *Each factor in (9) has rank at most $d_{\ell-1}$ or d_ℓ . In the quadratic scaling $n \asymp d_{\ell-1} d_\ell$, their Hadamard product can have a nontrivial bulk ESD (this is exactly the BP phenomenon).*

3.2 Tensorization (Hadamard vs Gram)

Define the tensor-product features

$$\omega_i^{(\ell)} := x_i^{(\ell-1)} \otimes \delta^{(\ell)}(x_i) \in \mathbb{R}^{d_{\ell-1}} \otimes \mathbb{R}^{d_\ell} \simeq \mathbb{R}^{d_{\ell-1} d_\ell}. \quad (10)$$

Then (9) is exactly the Gram matrix

$$K_{ij}^{(\ell)} = \frac{1}{d_{\ell-1} d_\ell} \langle \omega_i^{(\ell)}, \omega_j^{(\ell)} \rangle. \quad (11)$$

If we concatenate features across layers,

$$\omega_i := (\omega_i^{(1)}, \dots, \omega_i^{(L)}) \in \bigoplus_{\ell=1}^L \mathbb{R}^{d_{\ell-1} d_\ell},$$

then (up to deterministic layerwise constants that depend on the parametrization) the full NTK is a single sample covariance matrix:

$$\Theta_{ij}^{(L)} \approx \langle \omega_i, \omega_j \rangle. \quad (12)$$

Thus the deep problem is reduced to a *single* Bai–Zhou/Bai–Silverstein type theorem for the i.i.d. row vectors ω_i .

3.3 Why this matches the BP framework

In BP, the core object is the covariance tensor of ω_i (in our notation, $\mathbb{E}[\omega_i \otimes \omega_i]$ or its centered version). In the deep setting, the only new ingredient is that $\omega_i^{(\ell)}$ is built from deep features $x_i^{(\ell-1)}$ and deep backprop signals $\delta^{(\ell)}(x_i)$. The proof goals become:

1. show $\{\omega_i\}_{i=1}^n$ are independent across i and satisfy the quadratic-form concentration required by Bai–Zhou,
2. identify the limiting spectral law of the covariance operator of ω_i ,
3. apply the Marchenko–Pastur map with aspect ratio $n / \sum_\ell d_{\ell-1} d_\ell$.

4 Main theorem (deep quadratic scaling)

Assumption 1 (Quadratic scaling). *There exist constants $\gamma_1 > 0$ and $\gamma_{2,\ell} > 0$ such that as $n \rightarrow \infty$,*

$$\frac{n}{d_0 d_1} \rightarrow \gamma_1, \quad \frac{d_\ell}{d_{\ell-1}} \rightarrow \gamma_{2,\ell}, \quad \ell = 1, \dots, L.$$

Set $m := \sum_{\ell=1}^L d_{\ell-1} d_\ell$ and assume $n/m \rightarrow \gamma \in (0, \infty)$.

Assumption 2 (Data assumptions). *The entries of $X^{(0)}$ are i.i.d. centered with variance 1 and finite fourth moment. Moreover, for every $\varepsilon, \theta > 0$,*

$$\mathbb{P}\left(\left|\frac{1}{d_0} \|x_i^{(0)}\|_2^2 - 1\right| \geq \frac{d_0^\varepsilon}{\sqrt{d_0}}\right) \leq n^{-\theta},$$

where $x_i^{(0)}$ is the i -th row of $X^{(0)}$.

Assumption 3 (Activation assumptions). *The activation σ is locally Lipschitz and differentiable Lebesgue-a.e. Its derivative $\phi = \sigma'$ is pseudo-Lipschitz (in the sense used in BP) and has finite moments under $\mathcal{N}(0, 1)$. In particular, this covers smooth activations and can be extended to ReLU by approximation since the Gaussian mass at the kink is zero.*

Lemma 1 (Deep covariance tensor limit (main lemma)). *Assume Assumption 1, Assumption 2, and Assumption 3. Let $\omega = \omega(x) \in \mathbb{R}^m$ be the concatenated tensorized feature vector defined in (10)–(12), and let T be its conditional covariance operator defined in Section 5. Then, as $n \rightarrow \infty$:*

1. *for every deterministic operator B on $\mathbb{R}^m$ with $\|B\|_{\text{op}} \lesssim 1$,*

$$\mathbb{E}\left[(\omega^\top B \omega - \text{Tr}(BT))^2\right] = o(1) \cdot n^2,$$

2. *the empirical eigenvalue distribution of T converges in probability to a deterministic probability measure μ_{deep} on $[0, \infty)$.*

Theorem 1 (Deep quadratic-scaling NTK bulk law). *Assume Assumption 1, Assumption 2, and Assumption 3. Assume further that Lemma 1 holds. Then there exists a deterministic probability measure μ_{deep} on $[0, \infty)$ such that the empirical spectral distribution of the empirical NTK $\Theta^{(L)}$ converges weakly in probability to*

$$\mu_{\text{MP}}^\gamma \boxtimes \mu_{\text{deep}},$$

where μ_{MP}^γ is the Marchenko–Pastur law of shape γ and $\boxtimes$ is free multiplicative convolution.

4.1 Two standard tools from BP: rank bounds and the MP map

Lemma 2 (Stieltjes transform rank bound). *Let A, B be real symmetric $n \times n$ matrices. Then for any $z \in \{z \in \mathbb{C} : \Im z > 0\}$,*

$$\left|\frac{1}{n} \text{Tr}(A - z \text{Id})^{-1} - \frac{1}{n} \text{Tr}(B - z \text{Id})^{-1}\right| \leq \frac{\pi}{n \Im z} \text{rank}(A - B).$$

Proof. This is the standard bound obtained by comparing the empirical distribution functions of A and B and representing the Stieltjes transform by Lebesgue–Stieltjes integration by parts (see, e.g., Bai–Silverstein, Appendix A). $\square$

Definition 1 (Marchenko–Pastur map). *Let ν be a probability measure on $[0, \infty)$ and $\gamma > 0$. The Marchenko–Pastur map $\mu_{\text{MP}}^\gamma \boxtimes \nu$ is the free multiplicative convolution of ν with the Marchenko–Pastur law of shape γ .*

Theorem 2 (Bai–Zhou principle for Gram matrices, informal). *Let $\Omega \in \mathbb{R}^{n \times m}$ have i.i.d. rows with conditional covariance operator T (possibly random). Assume that for every deterministic $m \times m$ matrix B with $\|B\|_{\text{op}} \lesssim 1$, the quadratic form $\Omega_i^\top B \Omega_i$ concentrates around $\text{Tr}(BT)$ at the scale $o_{\mathbb{P}}(n)$, and that the empirical eigenvalue distribution of T converges to a deterministic limit μ_T . If additionally $n/m \rightarrow \gamma \in (0, \infty)$, then the empirical spectral distribution of $\frac{1}{m} \Omega \Omega^\top$ converges weakly to $\mu_{\text{MP}}^\gamma \boxtimes \mu_T$.*

Proof of Theorem 1

We follow the BP proof spine with minimal modifications.

Step 0: discard low-rank terms. By (6), $\Theta^{(L)} = K^{\text{CK}} + \sum_{\ell=1}^L K^{(\ell)}$. Since $\text{rank}(K^{\text{CK}}) \leq d_L$ and $d_L/n \rightarrow 0$, the rank bound for Stieltjes transforms (as in BP) implies K^{CK} does not affect the limiting ESD. Hence it suffices to study $\sum_{\ell=1}^L K^{(\ell)}$.

Step 1: tensorization. Define $\omega_i^{(\ell)}$ by (10) with $\delta^{(\ell)}(x_i)$ the i -th row of $\Delta^{(\ell)}$. Then (11) holds and by concatenation (12) gives a Gram representation of the full Hadamard part. Thus we may write

$$\sum_{\ell=1}^L K^{(\ell)} = \frac{1}{m} \Omega \Omega^\top,$$

where $\Omega \in \mathbb{R}^{n \times m}$ has rows ω_i .

Step 2: conditional Bai–Zhou (MP map). Conditioning on all weights $\{W^{(\ell)}\}_{\ell=1}^L$ and a , the rows $\omega_1, \dots, \omega_n$ are i.i.d. (as functions of the i.i.d. input rows). Let T be their conditional covariance operator on $\mathbb{R}^m$. Under the quadratic-form concentration condition of Bai–Zhou, the ESD of $\frac{1}{m} \Omega \Omega^\top$ converges (conditionally) to $\mu_{\text{MP}}^\gamma \boxtimes \pi$, where π is the limiting ESD of T .

Step 3: identification of the limiting covariance law. Lemma 1 provides the quadratic-form concentration required by Bai–Zhou and the convergence of the empirical eigenvalue distribution of the covariance operator T to μ_{deep} .

Conclusion. Combining Steps 0–3 yields the claimed limit $\mu_{\text{MP}}^\gamma \boxtimes \mu_{\text{deep}}$ for the ESD of $\Theta^{(L)}$. $\square$

5 Deep Gaussian equivalence and the covariance tensor

The two-layer BP analysis reduces the bulk law to the limiting spectral distribution of a *conditional covariance operator* (a tensor T), and then applies a Marchenko–Pastur map. In depth, the same reduction holds, but the tensor T is now built from the deep feature map $x \mapsto \omega(x)$. The new task is therefore to control the *fourth-order* fluctuations of this deep feature map in the quadratic scaling.

5.1 Conditional i.i.d. structure

Fix all weights $\{W^{(\ell)}\}_{\ell=1}^L$ and a . Since the input rows $x_i^{(0)}$ are i.i.d. and ω_i is a measurable function of $x_i^{(0)}$ and the fixed weights, the vectors $\omega_1, \dots, \omega_n$ are i.i.d. conditionally on the weights. This is the correct setting for the Bai–Zhou theorem, and it is exactly the conditional viewpoint used in BP for the two-layer model.

5.2 The deep covariance operator

Let $m = \sum_{\ell=1}^L d_{\ell-1}d_{\ell}$ and let $\omega(x) \in \mathbb{R}^m$ denote the concatenated tensor-product feature vector. Define the conditional mean and covariance (operators on $\mathbb{R}^m$)

$$\mathbf{E}[\omega] := \mathbb{E}[\omega(x_1^{(0)}) \mid \{W^{(\ell)}\}, a], \quad T := \mathbf{E}[(\omega - \mathbf{E}[\omega]) \otimes (\omega - \mathbf{E}[\omega])].$$

To apply Bai–Zhou, we need:

1. a quadratic-form concentration estimate for ω against deterministic test operators,
2. convergence of the empirical eigenvalue distribution of T to a deterministic law.

5.3 Where finite-width cumulants enter (precisely)

At this point the deep proof diverges from BP: for $\ell < L$, the backprop factor $U^{(\ell)}$ is sample-dependent, so T is not an explicit quadratic function of a single Wishart matrix as in the two-layer case. The required control is exactly a *connected four-point function* bound for the deep feature map (a fourth cumulant scaling as $O(1/d_{\ell-1})$ with a pairing structure in neural indices). This is the quantity computed recursively in the finite-width cumulant literature via diagrammatics.

5.4 A modular route to proving Lemma 1

The “deep upgrade” can be organized so that the BP engine (tensorization, resolvents, MP map) is unchanged. Only the *input module* (concentration properties of deep features) and the *tensor module* (the limiting covariance tensor) require new work.

A. Input module: preservation of concentration. The first step is to propagate concentration from $X^{(\ell-1)}$ to $X^{(\ell)} = \sigma(Z^{(\ell)})$. Conditionally on $X^{(\ell-1)}$, the columns of $Z^{(\ell)}$ are i.i.d. Gaussian, so empirical averages over neurons in layer ℓ concentrate at an exponential rate in d_{ℓ} . Since in the quadratic scaling $n \asymp d_0 d_1$ is only polynomial in the widths, a union bound over all samples (and all sample pairs) is compatible with these exponential tails.

B. Tensor module: fourth cumulants and pairing tensors. The second step is to identify the leading conditional covariance operator T of ω . In BP, this identification is obtained by explicit Gaussian integration by parts, producing a tensor Q with a pairing structure and a deterministic $O(1/d)$ correction. In depth, the corresponding object is the connected four-point function of the deep feature map $x \mapsto \omega(x)$. The finite-width NTK literature computes these connected four-point functions and shows they scale as $1/d_{\ell-1}$ with the same pairing structure in the neural indices. This is precisely the deformation size tolerated by the Bai–Zhou mechanism.

C. Tensor reduction (BP Theorem 4.2 style). Once the limiting tensor exists, the spectral bulk depends only on the limiting eigenvalue distribution of T . This step is universality-ready: deterministic $O(1/d)$ deformations of T do not change the limiting MP-map output beyond the order already absorbed by the BP error bookkeeping.

D. Resolvent/leave-one-out stability. One potential concern is that deep features are correlated across samples due to the shared weights. However, the BP/Bai–Zhou analysis is naturally formulated *conditionally on the weights*: given $\{W^{(\ell)}\}$ and a , the rows ω_i are i.i.d. because the input rows $x_i^{(0)}$ are i.i.d. Thus the leave-one-out expansions and martingale decompositions in the resolvent method run exactly as in the two-layer case, without requiring any additional “decay of correlation across samples” assumption.

Remark 4 (Status of Lemma 1). *In this note, Lemma 1 isolates the only genuinely new analytic input beyond BP. The modular route above shows that the remaining work is entirely in Steps A–B: prove concentration propagation for the deep forward/backward features at polynomial sample size $n \asymp d_0 d_1$, and import (or re-derive) the $1/d$ pairing structure for the connected four-point functions in a form that is uniform over such polynomially many inputs.*

Lemma 3 (Forward concentration by neuron averaging, informal). *Fix a layer $\ell \geq 1$ and condition on $X^{(\ell-1)}$. For $i, j \in \{1, \dots, n\}$ define the empirical forward Gram entry*

$$\widehat{\Sigma}_{ij}^{(\ell)} := \frac{1}{d_\ell} \left\langle x_i^{(\ell)}, x_j^{(\ell)} \right\rangle = \frac{1}{d_\ell} \sum_{k=1}^{d_\ell} \sigma(Z_{ik}^{(\ell)}) \sigma(Z_{jk}^{(\ell)}).$$

Then, for each fixed pair (i, j) ,

$$\widehat{\Sigma}_{ij}^{(\ell)} - \mathbb{E} \left[\sigma(G_1) \sigma(G_2) \mid X^{(\ell-1)} \right] = O_{\mathbb{P}} \left(d_\ell^{-1/2} \right),$$

where (G_1, G_2) is a centered Gaussian vector with

$$\mathbb{E}[G_1^2 \mid X^{(\ell-1)}] = \frac{1}{d_{\ell-1}} \left\| x_i^{(\ell-1)} \right\|^2, \quad \mathbb{E}[G_2^2 \mid X^{(\ell-1)}] = \frac{1}{d_{\ell-1}} \left\| x_j^{(\ell-1)} \right\|^2, \quad \mathbb{E}[G_1 G_2 \mid X^{(\ell-1)}] = \frac{1}{d_{\ell-1}} \left\langle x_i^{(\ell-1)}, x_j^{(\ell-1)} \right\rangle,$$

Proof. Conditionally on $X^{(\ell-1)}$, the columns of $Z^{(\ell)}$ are i.i.d. Gaussian, hence the summands $\sigma(Z_{ik}^{(\ell)}) \sigma(Z_{jk}^{(\ell)})$ are i.i.d. across k with finite variance under Assumption 3. The claim follows from a standard concentration inequality for averages of i.i.d. square-integrable random variables (e.g. Bernstein under sub-exponential tails, or Chebyshev under fourth moments). $\square$

Lemma 4 (Backward concentration by neuron averaging, informal). *Fix a layer $\ell \leq L - 1$ and condition on $X^{(\ell-1)}$ and $\Delta^{(\ell+1)}$. For $i, j \in \{1, \dots, n\}$ define the empirical backward Gram entry*

$$\widehat{\Pi}_{ij}^{(\ell)} := \frac{1}{d_\ell} \left\langle \delta^{(\ell)}(x_i), \delta^{(\ell)}(x_j) \right\rangle = \frac{1}{d_\ell} \sum_{k=1}^{d_\ell} \Delta_{ik}^{(\ell)} \Delta_{jk}^{(\ell)}.$$

Then, for each fixed (i, j) ,

$$\widehat{\Pi}_{ij}^{(\ell)} - \mathbb{E} \left[\phi(G_1) \phi(G_2) \mid X^{(\ell-1)} \right] \cdot \widehat{\Pi}_{ij}^{(\ell+1)} = O_{\mathbb{P}} \left(d_\ell^{-1/2} \right),$$

with (G_1, G_2) as in Lemma 3, and where $\widehat{\Pi}_{ij}^{(\ell+1)} := d_{\ell+1}^{-1} \langle \delta^{(\ell+1)}(x_i), \delta^{(\ell+1)}(x_j) \rangle$.

Proof. Under (5) we can write $\Delta^{(\ell)} = \phi(Z^{(\ell)}) \odot U^{(\ell)}$ with $U^{(\ell)} = \frac{1}{\sqrt{d_\ell}} \Delta^{(\ell+1)} W^{(\ell+1)\top}$. Conditionally on $(X^{(\ell-1)}, \Delta^{(\ell+1)})$, the columns of $\phi(Z^{(\ell)})$ are i.i.d. (since the columns of $Z^{(\ell)}$ are i.i.d. Gaussian) and the columns of $U^{(\ell)}$ are i.i.d. centered Gaussian with conditional covariance proportional to $\Delta^{(\ell+1)} \Delta^{(\ell+1)\top} / d_{\ell+1}$, and these two objects are independent because $W^{(\ell)}$ and $W^{(\ell+1)}$ are independent. Thus the summands in $\widehat{\Pi}_{ij}^{(\ell)}$ are i.i.d. across k with conditional mean $\mathbb{E}[\phi(G_1) \phi(G_2) \mid X^{(\ell-1)}] \cdot \widehat{\Pi}_{ij}^{(\ell+1)}$, and the same concentration argument as in Lemma 3 applies. $\square$

Completion of the proof under Lemma 1

Assuming Lemma 1, the Bai–Zhou theorem applies conditionally on the weights and yields the limiting ESD

$$\mu_{\text{MP}}^\gamma \boxtimes \mu_{\text{deep}}$$

for the Hadamard part of the NTK. The low-rank conjugate kernel is negligible by Lemma 2, which completes the proof of Theorem 1.